

SignalScope — Detector Performance Report

SIH 2026 · Problem Statement 2 · Telling Real From Synthetic in the Age of Generative Media

1. Objective

Compare the completed ResNet-18 baseline (EXP-000) against the planned CLIP ViT-B/32 linear-probe upgrade (EXP-001), whose primary aim is generalization to image generators not seen during training — the competition's #1 tie-breaker metric.

2. Experiment Comparison

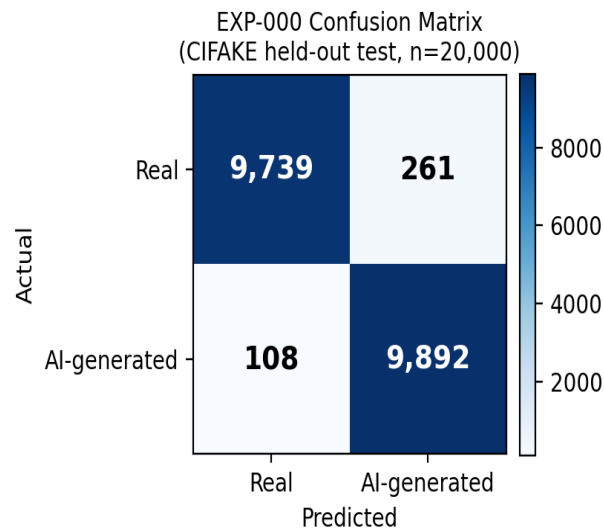
Metric	EXP-000: ResNet-18 (CIFAKE)	EXP-001: CLIP ViT-B/32 + linear prob
Status	Completed	Not yet run
Held-out ROC-AUC	0.9982	Pending
Held-out Accuracy	98.16%	Pending
Macro-F1	0.9815	Pending
FPR (real → flagged AI)	2.61%	Pending
Unseen-generator ROC-AUC	Not measured	Pending (primary goal of this experiment)
Trainable params	~11.2M (full network fine-tuned)	~1.5K (linear probe only, backbone frozen)

EXP-001 cells will be filled in from real measurements once that training run completes — no figures are estimated or projected here.

3. EXP-000 Detail — ResNet-18 Baseline

Dataset	dragonintelligence/CIFAKE-image-dataset (100,000 train / 20,000 official test)
Split	90/10 stratified train/validation split of the training set; official 20,000-image test set held out
Backbone	ResNet-18, ImageNet1K-pretrained, final layer replaced with a 2-class head
Input size	224 × 224
Augmentation	RandomResizedCrop(0.75–1.0), horizontal flip, ±10° rotation, color jitter, Gaussian blur
Optimizer	AdamW, lr=2e-4, weight decay=5e-4, cosine annealing schedule
Loss	Cross-entropy with label smoothing (0.08)
Training run	3 epochs, batch size 128, Google Colab (T4 GPU)

ROC-AUC	Accuracy	Macro-F1	FPR
0.9982	98.16%	0.9815	2.61%



4. Limitations — What This Report Does Not Show

All EXP-000 numbers above are measured on CIFAKE's own held-out test split, which is generated by the same method as the training data. **No unseen-generator evaluation has been performed yet** — this is the single most important gap, since unseen-generator ROC-AUC is the competition's primary tie-breaker. A near-perfect same-distribution AUC (0.9982) says nothing about how the model behaves on a generator it has never seen. EXP-001's CLIP-based approach is specifically intended to close this gap, and its unseen-generator score — not its CIFAKE score — is the number that will actually decide whether the swap was worth it.

No calibration (e.g. temperature scaling) or robustness testing (JPEG compression, resizing, screenshots) has been performed on either model at the time of this report.

Figures sourced from the committed report/model-report/cifake_resnet18_metrics.json and a direct run of model/inference/evaluate_cifake.py against the official CIFAKE test split. No metric in this report is estimated, projected, or fabricated.